IIT PATNA · AI/ML CAPSTONE · PROJECT SUBMISSION REPORT

NEURAL HEIST

A browser stealth-action game where you train your own AI teammates

Follows the 15-section Project Submission Report template of the IIT Patna B3 Capstone guidelines (Project 2: Build and Host a Video Game).

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Project title	Neural Heist – a browser stealth-action game where you train your own AI teammates
Capstone track	Project 2: Vibe Coding Challenge – Build and Host Your Own Online Video Game
Play online	https://reetika-k.github.io/neural-heist/
Project dashboard	https://reetika-k.github.io/neural-heist/dashboard.html
Repository	https://github.com/Reetika-k/neural-heist



Mission 1, Rooftop: the drone has marked the guards it recognised.

2. Project overview

Neural Heist is a 3D browser stealth game set in a neon Indian city. Each of its three missions needs one AI skill that the player trains in a **Lab** before playing: an image classifier for a scouting drone, a reinforcement-learning policy for a robot dog, and a search index for a hacking assistant backed by a real language model (Gemini). In the **Mission**, the teammate performs exactly as well as it was trained, and the guards are AI too (vision cones, suspicion, a state machine, A* pathfinding). An **AI Brain panel** shows every agent's real numbers live and exports them. The objective was to build a complete, tested, publicly hosted game with AI coding assistance, and to make AI/ML concepts tangible through play.

3. Problem statement

Students meet machine learning as formulas and accuracy numbers, but rarely feel what those numbers mean. That more and better labels make a classifier more reliable, that missing a guard and raising a false alarm are different mistakes, that a reward function decides what an agent learns, and that a language model can only answer from what it is given are ideas usually explained, not experienced. Most AI teaching tools are notebooks or quizzes, and most games that mention AI use it as decoration.

The target players are students aged 15 to 22 and casual gamers curious about AI, plus capstone evaluators who need to see the AI working. They play in a browser on a laptop or phone, in five-to-ten-minute missions, with no installation, account or paid service.

The challenge for the builder is to deliver a real game (objective, controls, scoring, levels, win and loss, pause and restart, sound, hosting) built with AI assistance, while making the AI inside it genuine: models trained by the player and used without faking, with measurable quality and visible consequences.

The proposed solution is Neural Heist. Before each mission the player trains one teammate. Label camera snapshots badly and the drone misses guards, whose vision cones then stay invisible; set it too trigger-happy and it flags civilians, drawing real guards over. Reward the robot dog for touching lasers and it walks straight into one, setting off the alarm. Build a poor search index and the assistant cannot work out the vault password. The score adds the teammate's measured quality to stealth and speed, and the AI Brain panel exposes every model's live numbers.

4. Target users

- Students (15–22) learning AI/ML who want intuition, not just definitions.
- Casual players who like short stealth or puzzle missions, on a laptop or a phone.
- Evaluators and teachers who want to see classification, reinforcement learning and retrieval-augmented generation behave live.
- Problem they face: AI concepts feel abstract; existing tools are passive; games rarely let you shape the AI.

5. Proposed solution

Need	How Neural Heist addresses it
Feel what labels and data quantity do	Mission 1 Lab: label 30 CCTV snapshots, train a k-nearest-neighbour classifier, compare training accuracy with accuracy on 40 held-back photos, get an overfitting warning with too few labels; the drone then uses that exact model
Feel precision vs recall	A "how sure before calling someone a guard" slider and a table of guards spotted, guards missed, false alarms and civilians cleared. In the mission a missed guard has no visible cone, and a false alarm draws a real guard over to check
Feel what rewards do	Mission 2 Lab: reward sliders (goal, laser, step), an episode count and ready-made settings; Q-learning trains live with a reward curve and policy arrows; the robot dog follows that exact policy. Vault extras: new random vaults, bonus batteries, blinking lasers, slippery puddles, a training challenge, and maps of what the dog learned and where it explored
Feel why retrieval matters for language models	Mission 3 Lab: chunk size, results per search (k) and a one-hop / two-hop switch; a "clues found" meter and a preview of the assistant's answer. In the mission, the retrieved emails go to Gemini, which works out the password; the player types it in
A real game	Stealth missions with a data drive, exit, timer, alarm, 3 stars, debrief, best scores, mission select, pause menu, restart, sound, settings, phone controls
Transparency	AI Brain panel with live values and JSON / CSV export; a project dashboard with every measurement

6. Key features

1. **Three missions, each with a Lab:** image classification (Rooftop), reinforcement learning (Laser Vault), retrieval-augmented generation (Executive Floor).
2. **AI guards:** 60° vision cones clipped by walls, suspicion meters, Patrol → Suspicious → Search → Chase state machine, A* pathfinding on the map grid; a guard walks over to investigate a false alarm.
3. **Teammates that use the player's real models:** drone (k-NN with the player's threshold), robot dog (greedy policy from the Q-table), hacking assistant (two-hop TF-IDF retrieval + Gemini 3.5 Flash-Lite through a Cloudflare Worker, with an offline fallback).
4. **Live language model, safely:** the Gemini key lives only on Cloudflare as a secret; the Worker accepts requests only from the game's site, limits each visitor to 20 requests an hour and rebuilds the prompt server-side.
5. **Scoring:** finish 1000 + stealth 500 + 10 points per second under par (at most 300) + up to 500 for the AI's measured quality; 3 stars; best score and stars per mission; total on the Heist Complete screen.
6. **Win and lose:** win with the drive at the exit; lose when caught, when the alarm runs out, when the dog trips a laser or gets lost, or after 3 wrong passwords. Restart or retrain in the Lab.
7. **Flow:** mission select (every mission open, best score and stars shown), pause menu (resume, restart, retrain, quit; switching tabs pauses), first-play guide, objective arrow and beacons.
8. **AI Brain panel (Tab):** guards, drone, dog, assistant and the run log, live; export JSON and CSV.
9. **Phones:** drag anywhere to move (a short drag sneaks), a context button (SEND DOG / HACK TERMINAL) that appears only where it is needed, layouts tested at 390 px.

10. **Look and sound:** low-poly people, drone and dog, lasers, neon signs (including Devanagari), bloom, illustrated CCTV portraits with a "Drone's view" toggle that shows the real 32×32 frames the classifier learns from; synthesised sound with a remembered mute; settings for difficulty, volume, graphics and colour-blind cones.

7. Solution architecture

See `docs/architecture.md` for the full diagram. In short:

```
Player → Browser game (Vite + Three.js), hosted on GitHub Pages
          ├── Lab: classifier.js · qlearning.js · rag.js        (models train in the browser)
          ├── Mission: movement · guards (FSM + A* + vision) · mission rules
          │            teammates use the trained models (drone · dog · assistant)
          ├── AI Brain panel: live values + JSON / CSV export
          └── Assistant ──POST /ask──▶ Cloudflare Worker ──▶ Gemini 3.5 Flash-Lite
                                (key secret · origin lock · 20 req/h · server-side prompt)
                          no answer ──▶ offline reasoning over the same retrieved emails
```

Every piece of game and AI logic is a pure JavaScript module with no Three.js or DOM inside, so it is unit-tested in Node; the 3D scene and screens only draw what those modules decide.

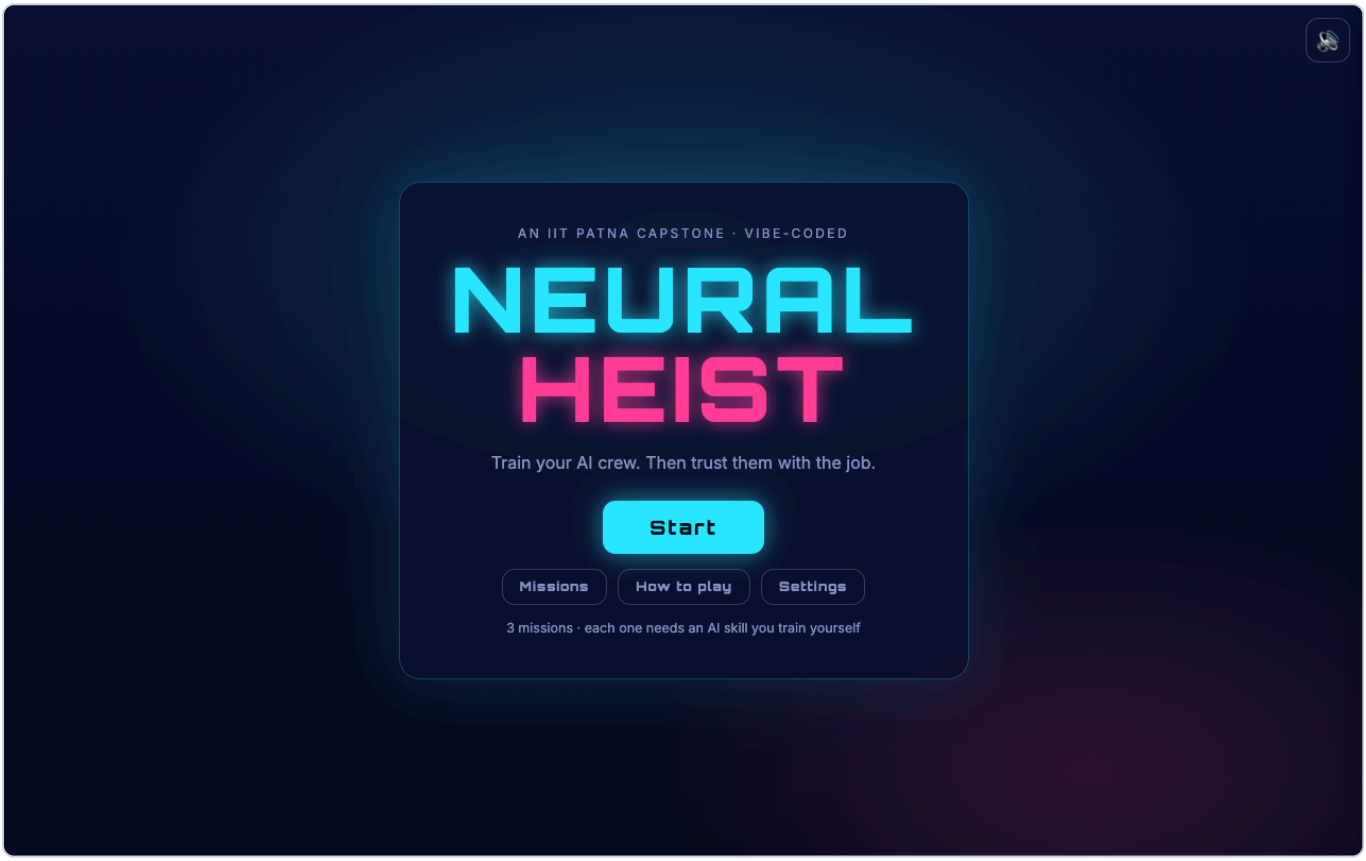
8. Tools and technologies

Part	Used	Note
Build tool	Vite 8	`npm run dev`, `npm run build`
3D	Three.js 0.186 (orthographic camera, instanced walls, bloom)	
Image classifier	k-NN (k = 3, distance-weighted vote, adjustable threshold) in plain JavaScript on 32×32 snapshots	the suggested TensorFlow.js MobileNet was replaced to avoid a 16 MB download and keep it testable
Reinforcement learning	Tabular Q-learning in plain JavaScript	as planned
Embeddings / retrieval	TF-IDF vectors, cosine similarity, two-hop retrieval	the suggested Transformers.js MiniLM was replaced for the same reasons
Language model	Gemini 3.5 Flash-Lite through a Cloudflare Worker (live); Ollama qwen2:7b locally	Groq as an optional backup in the Worker
Pathfinding and decisions	A* search, finite state machine, ray-cast vision	
Charts	Canvas	
Sound	Web Audio API synthesis	no downloads
Testing	`node:test`; Chrome DevTools Protocol scripts	no test-framework dependencies
Hosting	GitHub Pages, built, tested and published by GitHub Actions on every push to `main`; Cloudflare Workers for the language-model proxy	

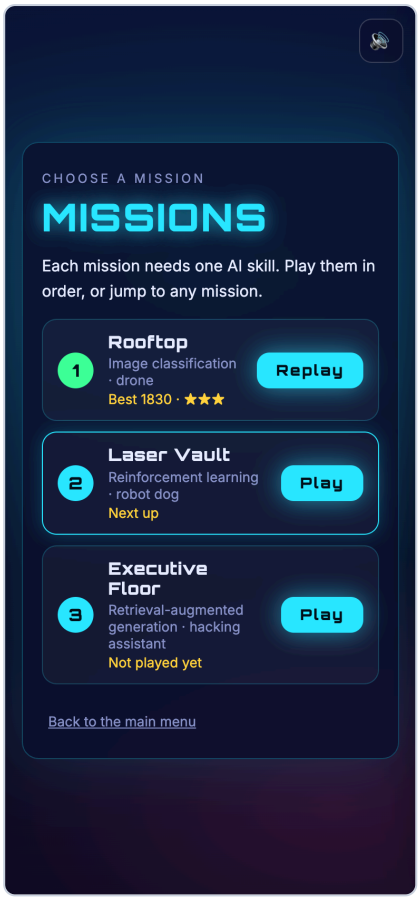
Part	Used	Note
AI coding assistant	vibe coding: prompts, code, tests and debugging	every prompt in `docs/prompts-log.md`

9. Working prototype

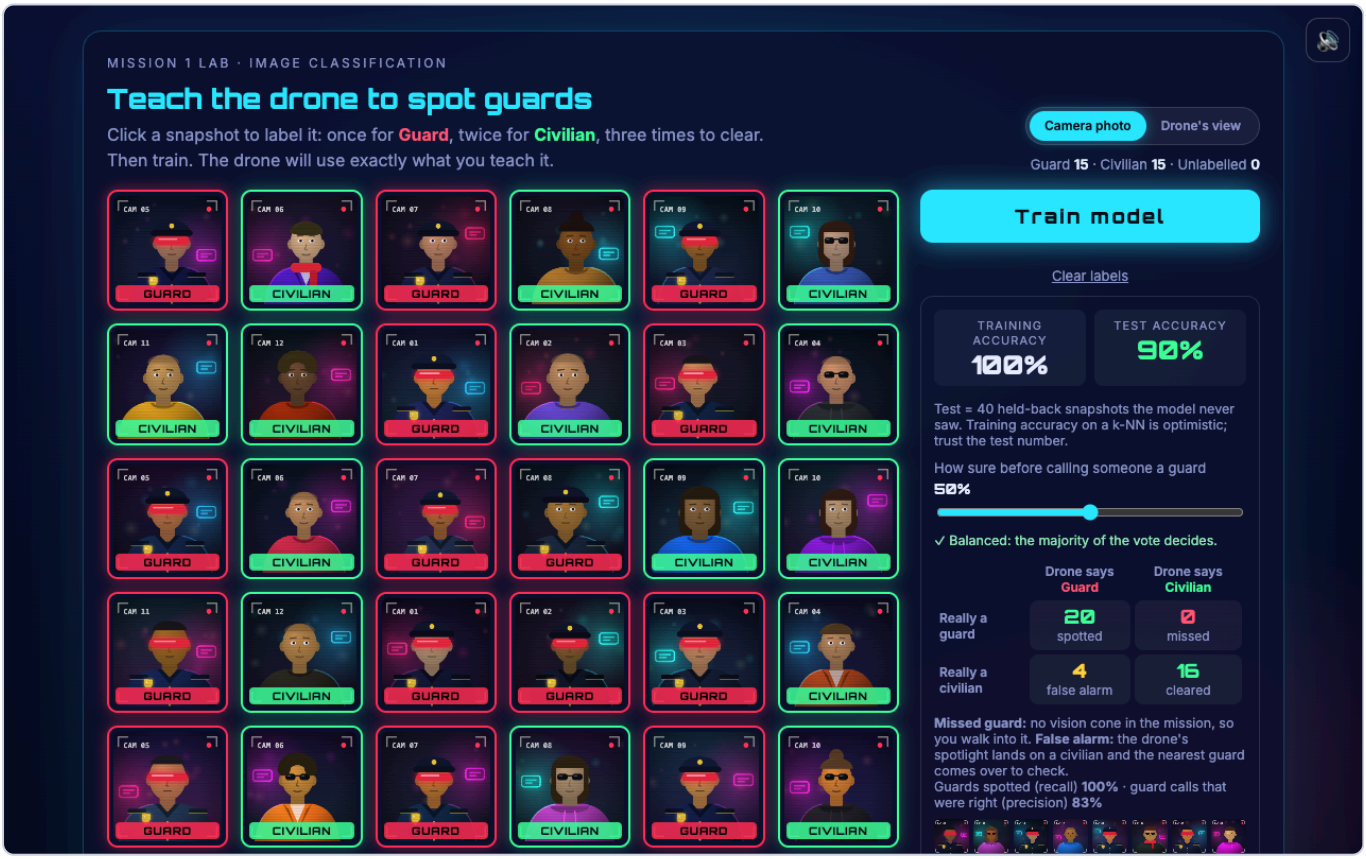
- **Play online:** https://reetika-k.github.io/neural-heist/ · **Dashboard:** https://reetika-k.github.io/neural-heist/dashboard.html
- **Run locally:** `npm install && npm run dev` → http://localhost:5173.
- **Mission 3 online** uses the live language model: the terminal shows "answered by gemini-3.5-flash-lite in … s". If the model is unreachable it reasons offline over the same emails, so the mission always works.



Start screen.



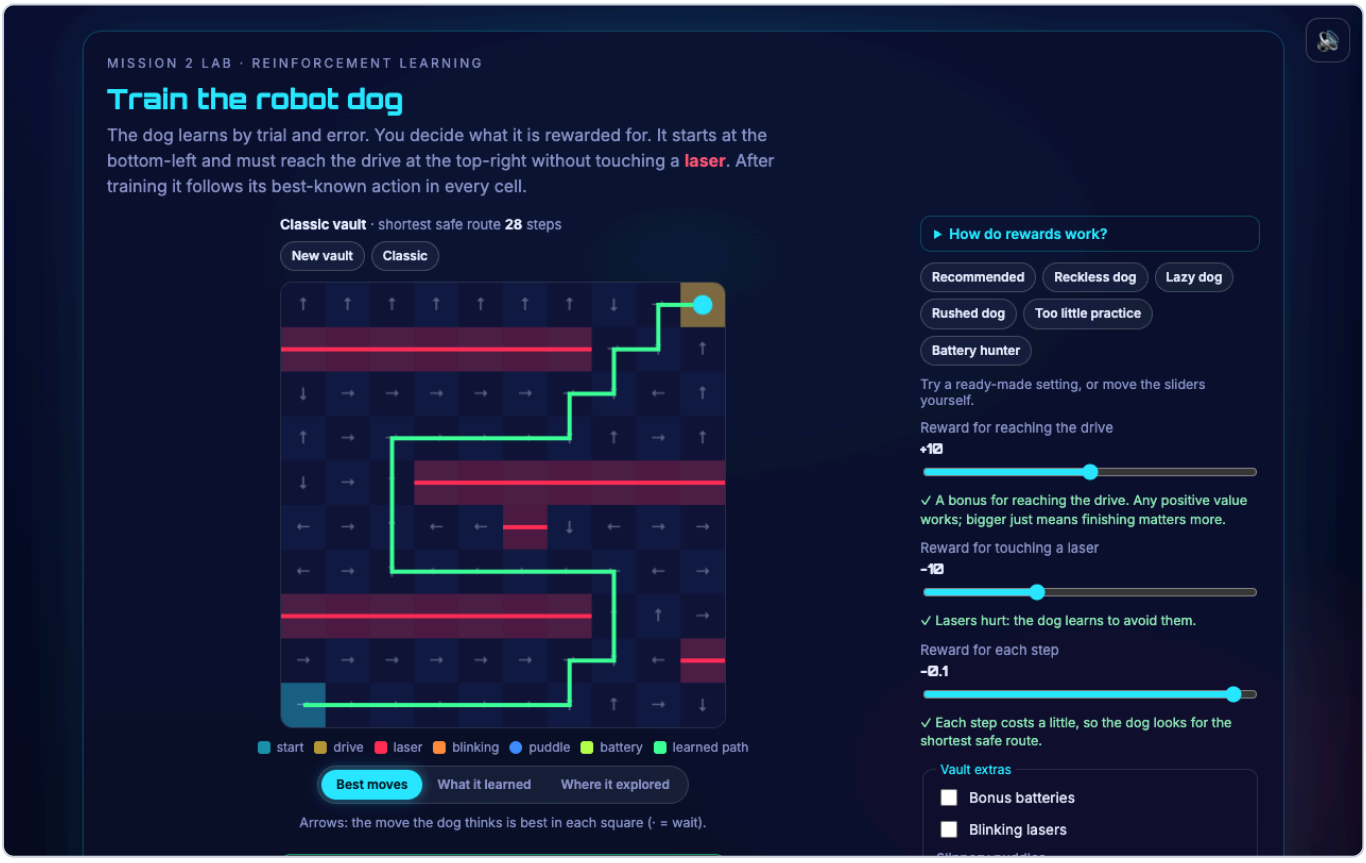
Mission select: every mission open, with best score and stars (phone layout).



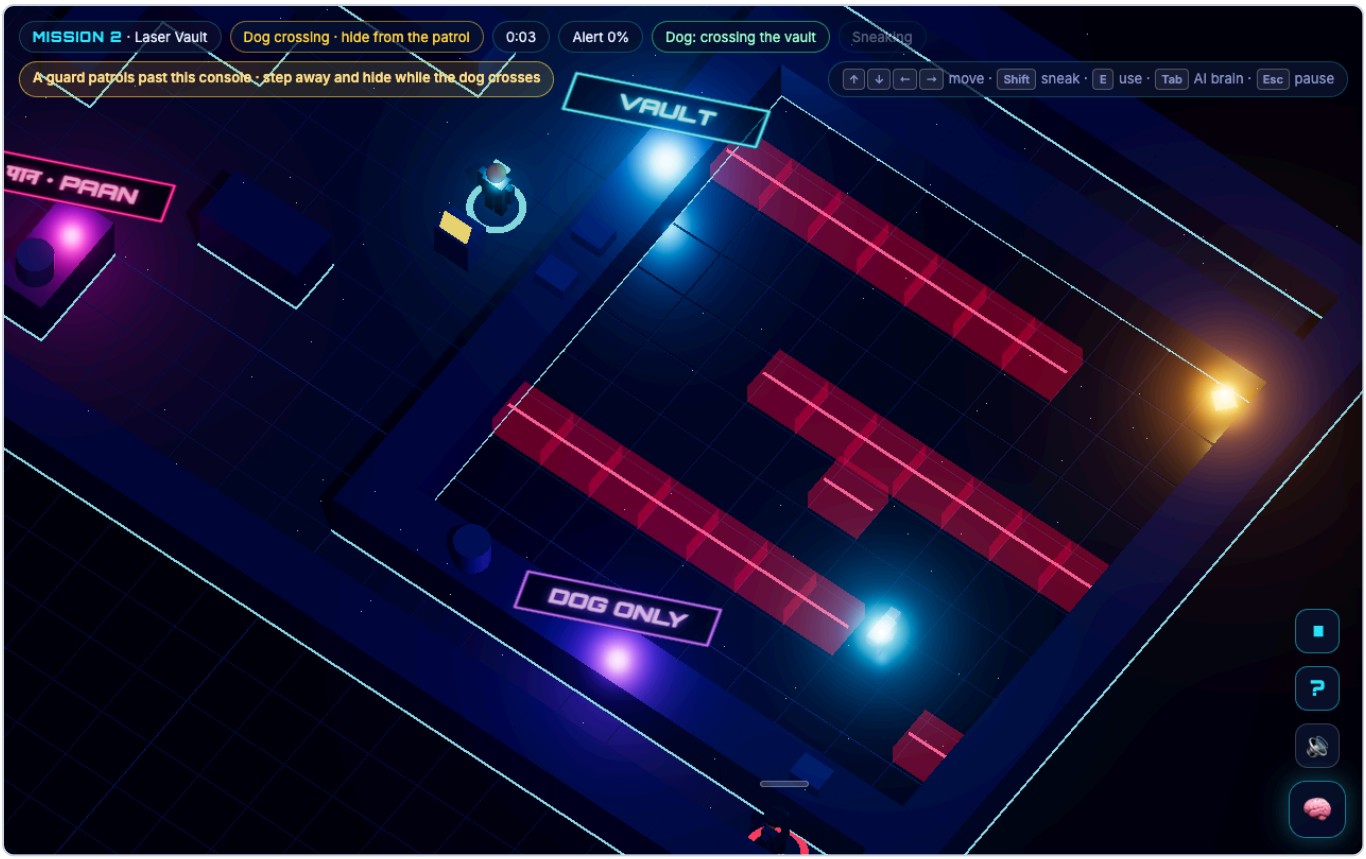
Mission 1 Lab: 30 labelled snapshots, test accuracy on 40 held-back photos, the "how sure before calling a guard" slider and the table of right and wrong calls.



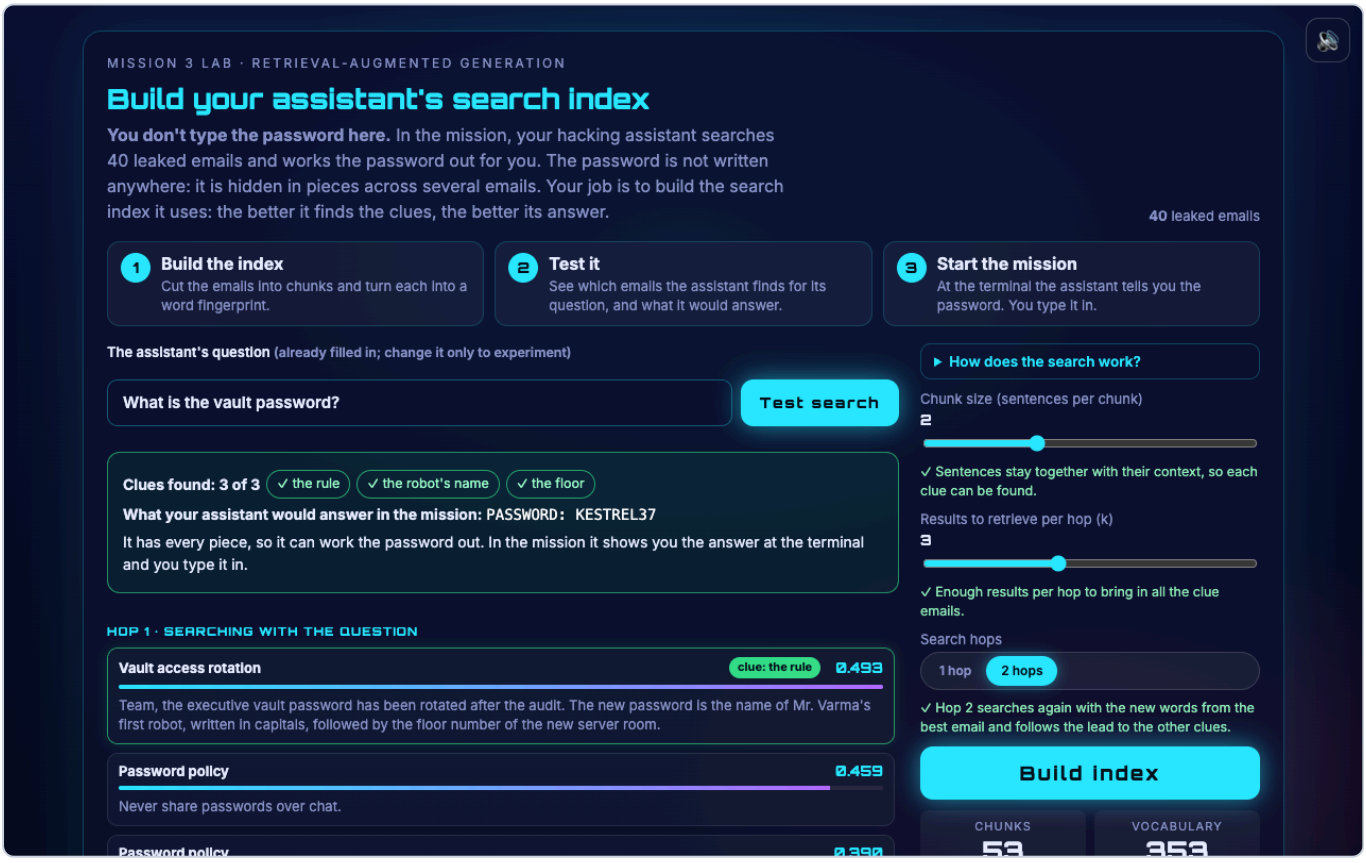
Mission 1: the drone tags the people it classified; guards it recognised show their vision cones.



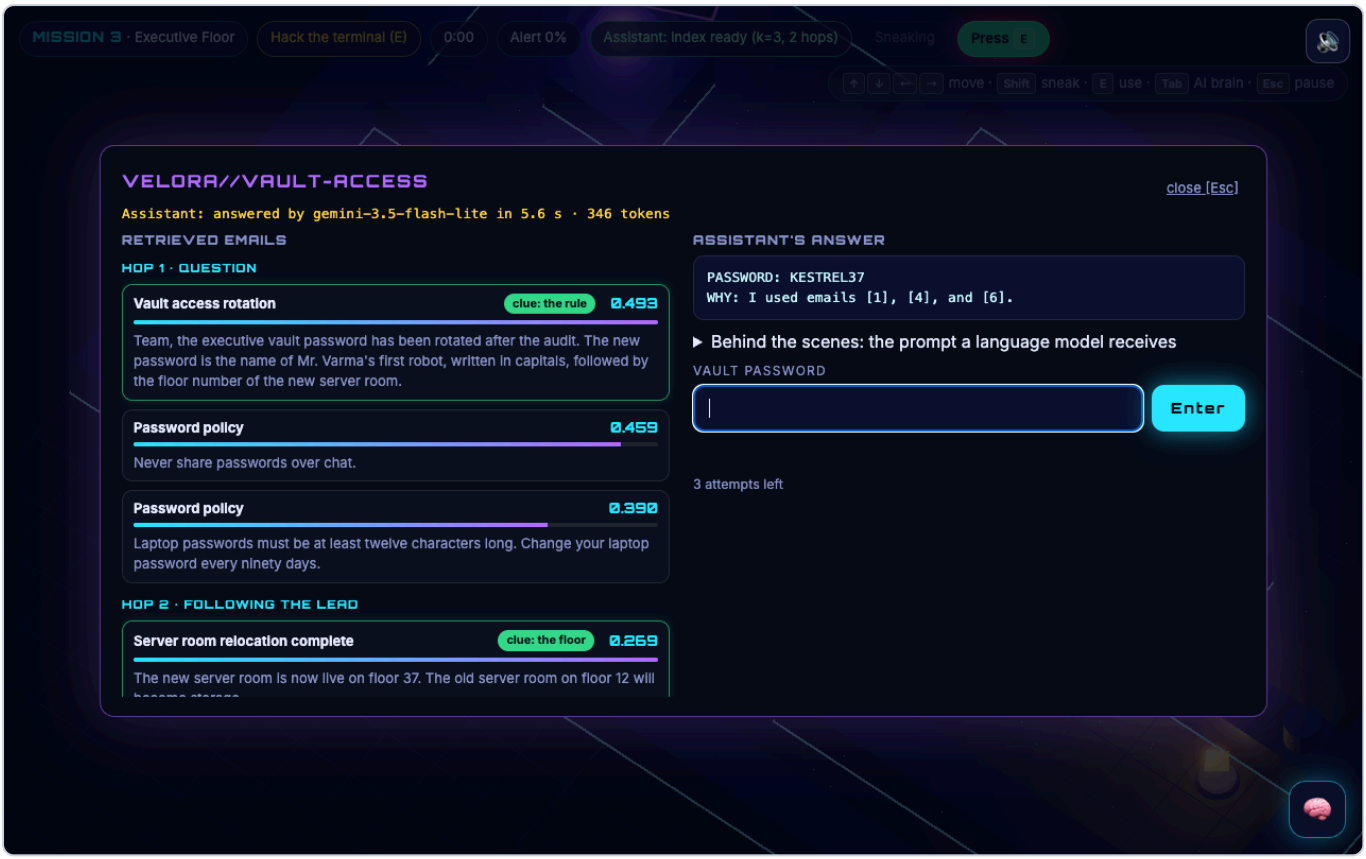
Mission 2 Lab: reward sliders, live Q-learning, the learned route.



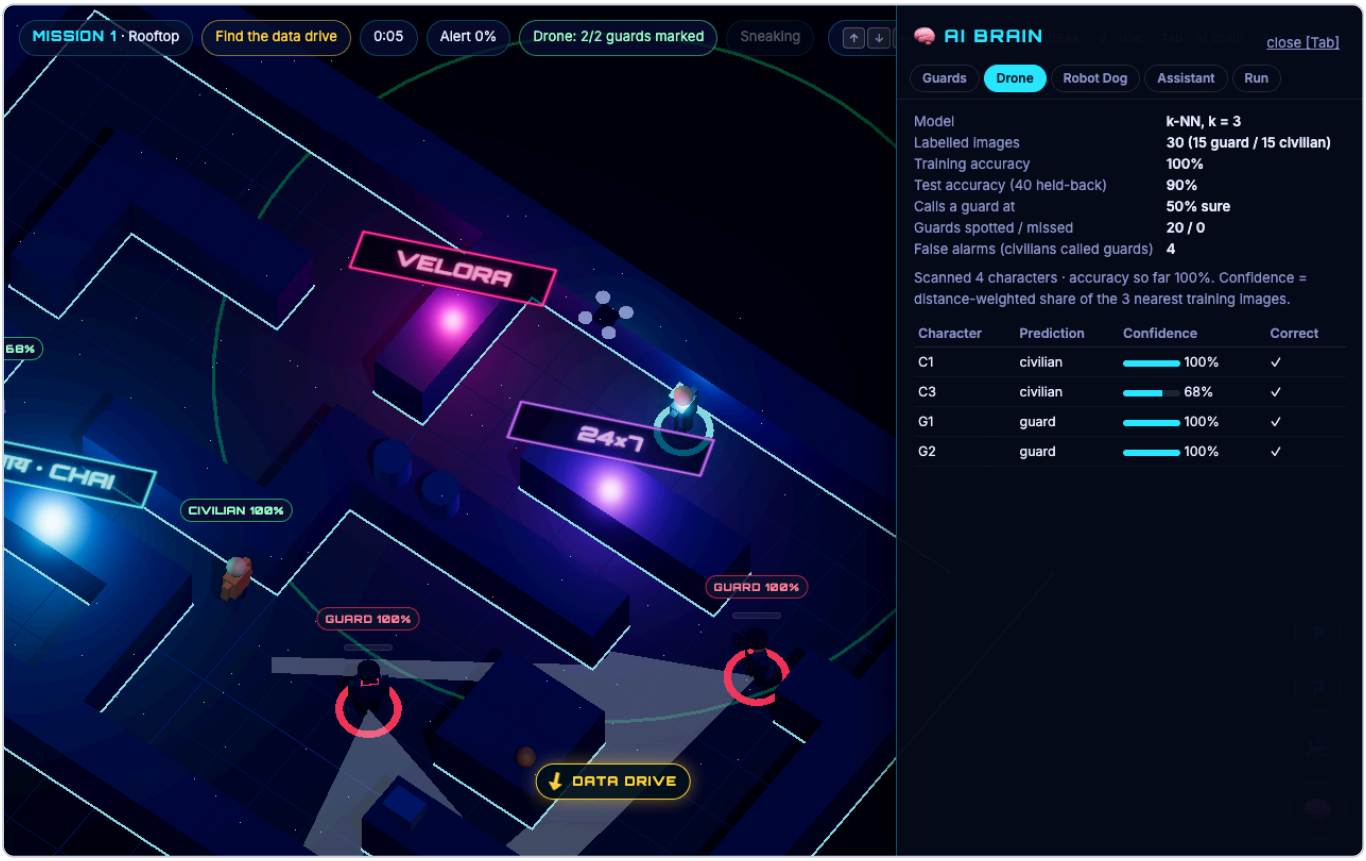
Mission 2: the robot dog crosses the laser vault with the policy it learned.



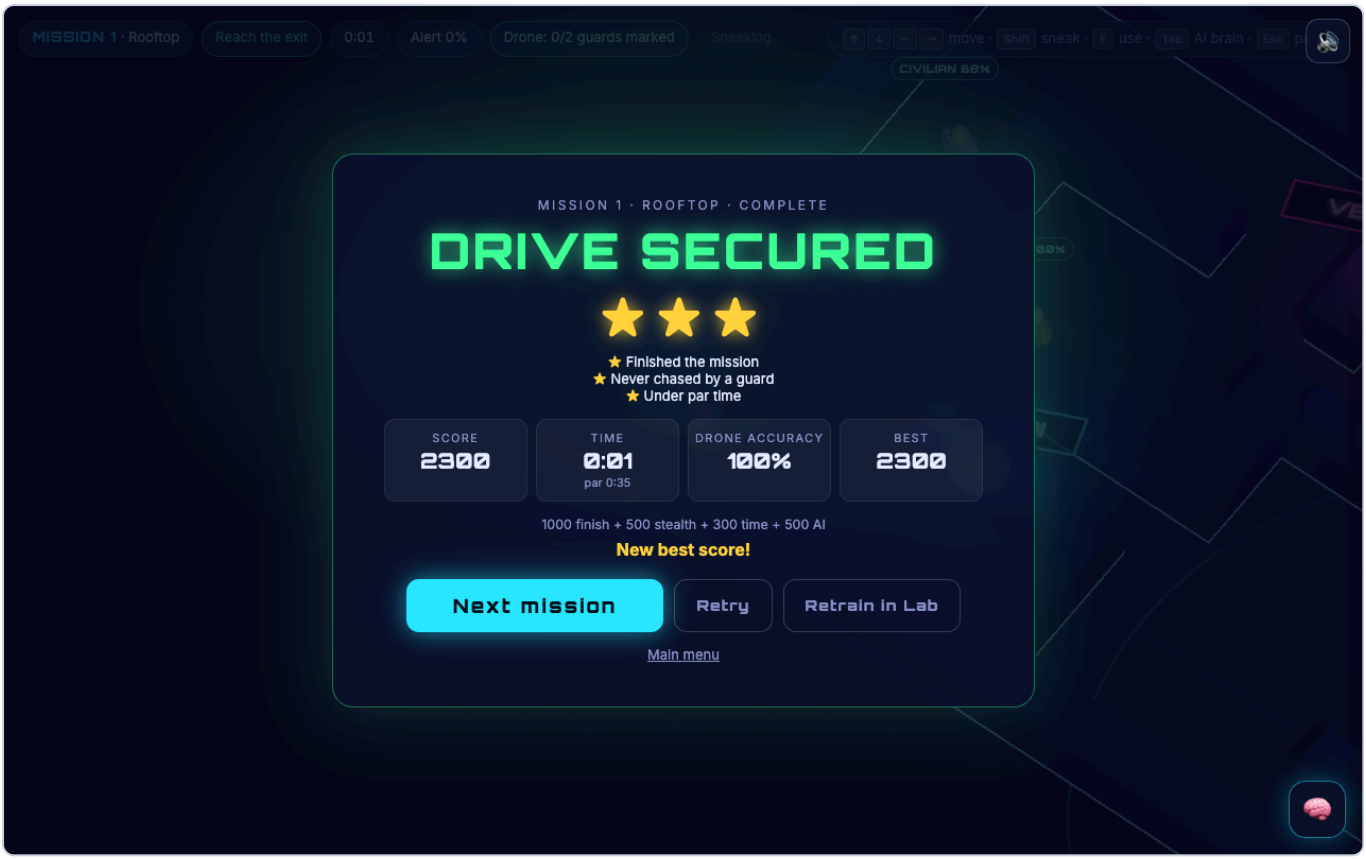
Mission 3 Lab: build the index, clues found 3 of 3, one-hop / two-hop switch, the answer the assistant would give.



Mission 3 on the live site: retrieved emails with clue badges, and the answer from Gemini 3.5 Flash-Lite.



AI Brain panel: live values from every AI agent, with export.



Debrief: score breakdown, stars and best score.

10. Prompts used

`docs/prompts-log.md` records every instruction given to the AI coding assistant, step by step from setup to deployment, with what worked, what failed and the follow-up prompt. Examples of how the game was shaped by prompts:

- *Building:* "Create a Lab screen for Mission 1. Show 30 camera snapshot images… Then a Train button trains a KNN classifier…"
- *Playtesting:* "I test neural heist with 1–3 users and there are some issues they face: 1. we are not saying how we are playing, the instructions are not clear 2. it is not mobile friendly" → the first-play guide, objective arrow, drag-to-move and the context action button.
- *Design review:* "check mission 1 to 3 if still any improvements are needed, act as an AI/ML developer and a video game developer" → ten measured improvements: mission select, pause menu, fair scoring, precision vs recall, false alarms with a cost, one vs two hops, and the live language model.
- *Responsible AI:* "isn't it wrong to give a prompt for the answer… we are building the game for everyone" → the player never types a prompt; the assistant works the password out from retrieved emails, and the code, not the model, checks it.

11. Testing report

`docs/test-report.md` lists all 12 README scenarios plus 37 more, each with expected and actual results. Current runs (2026-10-24):

Suite	Result
Unit tests (`npm test`): game rules, guards, A*, classifier, Q-learning, retrieval, Worker	77 / 77 pass
Browser test (`npm run test:e2e`): every screen, all three missions, Labs, Brain panel, export, settings, pause	88 / 88 pass on the dev server and on the live site with the language model connected
Score scenarios (`npm run test:scores`): every debrief checked against the scoring rules	38 / 38 pass (dev and live)
Phone check (`npm run test:phone`), 390 × 844	15 / 15 pass (dev and live)

Key measured results (all reproducible with fixed seeds, collected in the dashboard):

What	Result
Classifier, 30 correct labels	90 % on the Lab's 40 held-back photos; 90 % on 400 unseen; 62 % with only 4 labels
Threshold trade-off (400 unseen)	30 % sure: 0 guards missed, 67 false alarms · 50 %: 0 missed, 39 false alarms · 70 %: 11 missed, 22 false alarms
Q-learning, sensible rewards	shortest 28-step safe route on 20 / 20 seeds; a positive laser reward hits a laser on 20 / 20; no step penalty gets lost on 20 / 20
Retrieval recall of the 3 clue emails	3 / 3 with chunks of 2+ sentences, k ≥ 3 and two hops; 1 / 3 with one-sentence chunks; 1 / 3 with one hop at every setting
Live language model	answers KESTREL37 in 1.4–5.6 s; requests from other websites are refused (403)

`reports/dashboard.html` collects every measurement and is rebuilt from the game's own code by `npm run dashboard` . `docs/REQUIREMENT_MAPPING.md` maps every requirement to its implementation and test.

12. Challenges and solutions

`docs/challenges.md` lists 33 problems and fixes, each with what broke, the cause, how AI helped and the fix. A selection:

Problem	Cause	Fix
The assistant could not find the password even with a good index	the sentence splitter broke at "Mr.", separating the key clue from its subject	found by measuring recall and reading the chunks; abbreviations handled
A perfectly trained drone scored only 60 %	two look-alike civilians among the Rooftop characters	fairer character photos chosen by measurement: perfect training now scores 100 %
Rushing beat good training	uncapped time bonus far larger than the AI points	tighter pars, 10 points per second, capped at 300
Playtesters did not know how to play; not usable on phones	no guidance; keyboard-only design	first-play guide, objective arrow and beacons; drag-to-move and a context button
The game crawled on slow machines	a frame-time cap meant to stop wall tunnelling	fixed-size physics substeps

Problem	Cause	Fix
Live Gemini returned 404	the configured model had been shut down by Google	switched to gemini-3.5-flash-lite after checking the current model list

13. Final outcome

A complete, publicly hosted three-mission game with real, player-trained AI and a live language model, tested end to end in a real browser on the development build and on the live site, on desktop and phone sizes. The player trains a classifier, a reinforcement-learning agent and a retrieval index, and sees each one succeed or fail in play, with the numbers to explain why.

Key learnings: build game logic as pure, testable modules; measure ML behaviour before building UI around it; read the data when a metric looks wrong; keep secrets on a server, never in a public web page; never let the language model make the decision that matters (the player types the password, the code checks it); and test the live site, not only the dev server.

14. Future enhancements

New missions: clustering (a server farm), regression (guard patrol timing), a small neural network (a sound lock) and a final mission against a company AI that adapts security to the player's style. Also: TensorFlow.js MobileNet and neural embeddings as an optional mode, a leaderboard, recorded audio and modelled assets, and a strict global rate limit with Workers KV.

15. Project demo

- Five-minute demo script: `docs/DEMO_SCRIPT.md`.
- Play online: https://reetika-k.github.io/neural-heist/
- Project dashboard: https://reetika-k.github.io/neural-heist/dashboard.html
- Repository: https://github.com/Reetika-k/neural-heist